

SlimSeg-Net: Real-Time Medical Image Segmentation

Real-time segmentation networks trade accuracy for speed. We introduce SlimSeg-Net, which uses Ghost convolutions in the encoder to cut FLOPs and a Cross-Attention Gate that fuses spatial and channel context before the decoder. On the KP-MRI benchmark SlimSeg-Net reaches 0.894 Dice with only 2.1M parameters, and runs at 42 FPS on a consumer GPU, surpassing larger baselines on the best accuracy-per-parameter frontier. The attention module transfers to chest X-ray pathology masks with negligible retraining cost. An ablation shows the Cross-Attention Gate contributes +1.9 Dice over a Ghost-Conv baseline. We evaluate against six baselines across three data splits. In the discussion we note that attention was tuned for 2D slices and that 3D volumetric and multi-center generalization are not yet validated. Existing backbones are either heavy or underperform. Clinical demand requires edge-deployable models, so a lightweight yet accurate network is needed. We also report Dice, IoU, parameters, FLOPs, and FPS for every model. The gate was tuned only for 2D slices, which is the main limitation of our study.